Mathematical review of a stipulated compactification

# Volume inflection with a dynamical dilaton

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## 1. What the comparisons can exclude

In five-dimensional Einstein proper time write $H = \dot{a}/a$, $K = \dot{b}/b$, $\theta = 3H + K$, $x = \log(a/a_*)$, and $R = \log(V/V_*)$ for $V = 2\pi a^3 b$. The classical tuned solution has $R = -24x^3 + 270x^4 + O(x^5)$. Its stress is prescribed. The three submitted manuscripts compare this solution with several restricted string-motivated calculations.

**Vacuum.** At tree level and two derivatives, the unexcited flux-free toroidal reduction with fixed flat spectators and constant dilaton gives vacuum Einstein equations [1]. For $H > 0$, the constraint forces $K = -H$. Hence $R = 2x$. From $x = -0.01$ to $+0.01$, the vacuum log-volume increases by $0.04$; the tuned history decreases by $48.67431$--$48.67432$ parts per million. The predicted changes are separated by at least $0.04004867431$. Uniform errors below $0.01001216$ on each of two readings keep their allowed change intervals disjoint. To resolve the smaller decrease against zero from those readings alone requires errors below about $24.337$ parts per million. The vacuum calculation has no line emitter.

**Twisted pair.** The effective static sum for one matched pair of light degrees of freedom gives $-(31/16)C_L$, with $C_L = 3\zeta(5)/(4\pi^2 L^5)$ and $L = 2\pi b$. The fermion twist changes the winding sum by $-15/16$ relative to the periodic series. The coefficient of a complete string partition function depends on its spectrum and boundary conditions [2]. For the isolated density $-D = -C/b^5$, external pressure $p_x = -\rho$ and circle pressure $p_y = 4\rho$, $\ddot{V}/V = -\kappa D < 0$. Any stationary volume is a strict maximum.

**Positive constant.** Under the five-dimensional Einstein rescaling, the stated dilaton-independent genus-one string-frame density becomes $A - D$, where $A = \Lambda_s e^{10\phi/3}$, $D = C/b^5$, and $b$ is now the Einstein-frame radius. At fixed $b$, $\partial_\phi(A - D) = 10A/3 > 0$. Freezing the dilaton over an interval fails its equation with only these sources. Published rolling solutions have additional model-specific field content and initial conditions [3]. The following local calculation tests what happens when the dilaton in this particular ansatz is allowed to obey its own equation.

## 2. Cubic agreement forces quartic disagreement

Use the standard two-derivative Einstein kinetic term $-(4/3)(\partial\phi)^2/(2\kappa)$ and the source $A-D$ above [1]. Assume three equal external rates, fixed spectators, no additional forces or corrections. This defines a reduced test model. The equations for microscopic spectators and a derivation of the two terms from one string compactification are outside it.

Put $T = 2\dot{\phi}^2/(3\kappa)$. The density and pressures are $\rho = T+A-D$, $p_x = T-A+D$, $p_y = T-A-4D$. Therefore

$$3H(H+K) = \kappa(T+A-D), \qquad \dot{\theta}+\theta^2 = \kappa J/3, \quad J = 8A-3D,$$
$$\ddot{\phi}+\theta\dot{\phi} = -5\kappa A/2, \qquad \dot{A} = (10/3)A\dot{\phi}, \quad \dot{D} = -5KD.$$

At a stationary event with $H > 0$ and $\ddot{V} = 0$, these equations give $K = -3H$, $D = 8A/3$. Write $v = \dot{\phi}/H$ at that event. The constraint requires $\kappa A/H^2 = 2(9+v^2)/5$. For the coefficient $c_3$ in $R = c_3x^3 + c_4x^4 + O(x^5)$ one obtains

$$c_3 = \frac{\kappa\dot{J}}{18H^3} = \frac{8}{27}(9+v^2)(2v-9), \qquad c_3+24 = \frac{8}{27}v(2v^2-9v+18).$$

The quadratic factor has negative discriminant. Thus matching the classical $c_3 = -24$ requires $v = 0$. This is admissible instantaneous rest: choose $A_* > 0$, fix $b_*$ by $C/b_*^5 = 8A_*/3$ and $H_*^2 = 5\kappa A_*/18$. The regular evolution equations propagate the constraint and admit a local solution. At the event $\ddot{\phi}_* = -9H_*^2$.

The quartic differs. At $v = 0$, $H, K, \dot{H}, \dot{K}, A, D, \dot{J}$ agree with the classical data having a constant positive vacuum density $A_*$. The only new contribution to $\ddot{J}$ at that instant is $8\ddot{A} = 8(10/3)A\ddot{\phi}$. Since

$$24c_4 = \frac{\dddot{\theta}}{H^4} - \frac{6\dot{H}\ddot{\theta}}{H^5},$$

the fourth derivative shifts by $\kappa\, 8(10/3)A\ddot{\phi}/(3H^4) = -288$. Hence $c_4 = 270 - 288/24 = 258$. Matching both coefficients is impossible in this reduced system. Numerical integration gives a two-epoch drop of about $48.6150$ parts per million for this local solution. That estimate is a diagnostic; the coefficient and exclusion are exact. The embedded verifier also differentiates the full spatial equations directly.

## 3. Comparing quantities under circle duality

Write $s = b_s/\sqrt{\alpha'}$ for the string-frame radius in string units. With a spacelike isometry, zero mixed components and fixed string-frame spectators, the leading Buscher transformation sends $s \mapsto 1/s$ and $\phi \mapsto \phi - \log s$ [1,4]. In the five-dimensional Einstein frame this gives

$$\frac{\tilde{a}}{a} = s^{2/3}, \quad \frac{\tilde{b}}{b} = s^{-4/3}, \quad \frac{d\tilde{t}}{dt} = s^{2/3}, \quad \frac{\widetilde{V}}{V} = s^{2/3}.$$

At $\theta = \dot{\phi} = 0$, $K = -3H$, the transformed expansion is $-2Hs^{-2/3}$. This tests the transformation of a geometric quantity. A claim of equivalent string solutions would additionally require dualizing the sources, twists and probes in a genuine construction.

The reduced four-dimensional Einstein metric $b\, g_{E,\text{ext}}$ is invariant up to common fixed normalization. Its external scale is $a_4 \propto a\sqrt{b}$, so at corresponding epochs

$$\log(a_4/a_{4*}) = (R - x)/2.$$

In the stated vacuum it expands; at the tuned inflection it contracts. Across the two stated epochs its log-scale change is $+0.01$ in the vacuum and approximately $-0.0100243372$ in the tuned model. The signs survive a change to four-dimensional Einstein proper time, which increases with five-dimensional proper time. Epochs must be matched by physical clocks or transformed probes. A fixed numerical $x$ need not label the same event after duality.

There is another invariant, the reduced string-frame spatial measure $\mathcal{M} = e^{-2\phi} a_s^3 b_s \propto e^{2\phi/3} V$. It illustrates the consequence of scalar acceleration. On the cubic-matched rolling solution of section 2, its logarithmic slope vanishes at the event, but its second proper-time derivative is $-6H_*^2$. The invariant measure has a strict local maximum while the bare volume has a stationary inflection. Neither quantity comes with a calibrated physical detector in these models.

**Scope of the verification.** The submitted script reproduces the $31/16$ coefficient, frame transformations and normalized tree-amplitude angular ratios $3/4$ and $13/16$. The latter ratios have limited identifying power: for $s+t+u=0$, symmetric polynomials of degrees three and five are each fixed up to scale, by $stu$ and $(st+su+tu)stu$, respectively. The pole spectrum and normalization contain additional information. The three manuscripts supply no matched cosmological observation or verified string embedding.

[1] D. Tong, Lectures on string theory, secs. 7--8.

[2] M. Borunda, M. Serone and M. Trapletti, On the quantum stability of IIB orbifolds and orientifolds with Scherk--Schwarz SUSY breaking.

[3] T. Coudarchet, C. Fleming and H. Partouche, Quantum no-scale regimes in string theory.

[4] A. Giveon, M. Porrati and E. Rabinovici, Target space duality in string theory.